
3AL Tutorial 8 2024

1. Let X be the set of subgroups of a group G . Prove that conjugation is a group action on X .
2. Let G be a group with subgroup H . Prove that $H \triangleleft G$ if and only if H is a union of conjugacy classes.
3. Let H be a subgroup of G and $N_G(H)$ be the normaliser of H . Prove the following:
 - (a) $N_G(H)$ is a subgroup of G .
 - (b) $H \triangleleft N_G(H)$.
 - (c) If K is a subgroup of G such that $H \triangleleft K$, then $K \subseteq N_G(H)$.
 - (d) $H \triangleleft G$ if and only if $N_G(H) = G$.

Then justify the statement “The normaliser of H is the largest subgroup of G that contains H as a normal subgroup”.

4. Let G be a group and H a subgroup of G . The **centraliser** of the subgroup H is defined as $C_G(H) = \{g \in G : gh = hg \text{ for all } h \in H\}$. Prove that $C_G(H) \triangleleft N_G(H)$.
5. Let G be a group with subgroups H and K . Prove that if H and K are conjugates then $N_G(H)$ and $N_G(K)$ are conjugates.

Extra problems

- (I) Let G be a finite group with exactly two conjugacy classes. Show that $|G| = 2$.
- (II) Prove that if G is a finite group and $a \in G$ such that $|\text{class } a| = 2$, then G is *not* simple.

NOTE: Recall (from Tut 5 Extra Problems) that a group is called simple if its only normal subgroups are itself and the trivial subgroup.